

Nicholas Brar

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Education

Stony Brook University , Mathematics	Stony Brook, NY
- Member of Stony Brook Chess Club 2nd place in the Collegiate Chess League in Fall 2024.	2021 – 2025

Experience

EcoDataLab , Climate Data & Modeling Intern	Berkeley, CA (Remote)
- Redesigned the Consumption Based Emissions Inventory (CBEI) modeling pipeline using CEX data, replacing a single model with 32 annual models and improving average predictive accuracy (R^2) by 0.08 - Built a unified React map component that replaced iframe pages, cut load times by 5x, and enabled fully interactive maps for emissions by tract - Built an end-to-end R workflow combining supply-chain and economic data to compute and visualize national non-GHG impact factors at the local level - Automated the conversion of high-level impact data into detailed sector-level outputs, enabling more precise and locally relevant analysis	Sept 2025 – Feb 2026 6 months
Dealer Shock , Software Engineer Intern	Cleveland, OH (Remote)
- Built the company’s official React Native mobile application from scratch using Expo, pushing development 12 months ahead of its original estimated release date - Automated VIN scanning for clients, reducing manual entry by 90% - Collaborated in 2-week sprints using Agile methodologies to deliver iterative improvements	Jan 2025 – May 2025 5 months
Stony Brook University Mathematics Department , DRP Undergraduate Scholar	Stony Brook, NY
- Built a Python-based MNIST classification pipeline using topological features, surpassing prior topological data analysis (TDA) benchmarks - Applied modern software design patterns to parallelize feature extraction across 13 branches to extract rich and discriminative feature sets - TDA publication showcased on Stony Brook’s DRP site	Dec 2024 – May 2025 6 months

Skills

- Languages
- ML Frameworks & Libraries
- Developer Tools

Languages

- English
Native
- Spanish
Basic
- Hindi
Basic

Interests

- Climate & Environment
- Machine Learning

Certificates

Advanced NLP with Python for Machine Learning

Sept 2025

Java (Basic)

July 2024

Projects

Fast American Option Pricing

Python, C++, Quantitative Finance, Numerical Methods

- Implemented a high-performance spectral collocation method for American option pricing in C++, based on Jacobi–Newton iteration, Gauss–Legendre quadrature, and Chebyshev polynomial interpolation
- Reproduced results from research literature, achieving computational throughput of 100,000 option prices per second per CPU in the Black–Scholes model

Vehicle Miles Traveled (VMT) Model

Python, Pandas, NumPy, scikit-learn, Random Forest

- Built an end-to-end data pipeline to clean, integrate, and feature-engineer household, vehicle, and person-level survey data (NHTS) for predictive modeling
- Developed and evaluated regression and ensemble models (Linear Regression, Random Forest) to predict household vehicle miles traveled (VMT), with weighted performance metrics and feature importance analysis